

Discrete Mathematics

Week 4: Predicate Logic - Practice & Examples

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Today's Practice Session

We will practice:

- 1 Propositional Functions & Predicates
- 2 Universal and Existential Quantifiers
- 3 Translating English to Predicate Logic
- 4 Translating Predicate Logic to English
- 5 De Morgan's Laws for Quantifiers
- 6 Nested Quantifiers
- 7 Domain-dependent Translations
- 8 Computer Science Applications

Multiple examples with detailed solutions for each topic!

Section 1: Propositional Functions - Examples

Example 1: Let $P(x)$ denote " $x^2 < 20$ "

Domain: All positive integers

Evaluate:

- $P(3) = ?$
- $P(5) = ?$
- $P(4) = ?$

Solution: Propositional Functions

Let $P(x)$ denote " $x^2 < 20$ ", Domain: All positive integers

- $P(3): 3^2 < 20 \rightarrow 9 < 20$ TRUE
- $P(5): 5^2 < 20 \rightarrow 25 < 20$ FALSE
- $P(4): 4^2 < 20 \rightarrow 16 < 20$ TRUE

Question: For which positive integer values is $P(x)$ true?

Answer: $x \in \{1, 2, 3, 4\}$

Propositional Functions - CS Example

Example 2: Let $Q(n)$ denote “Algorithm A runs in $O(n)$ time on input size n ”

Evaluate:

- $Q(\text{Merge Sort}) = ?$
- $Q(\text{Linear Search}) = ?$
- $Q(\text{Binary Search}) = ?$

Solution: CS Example

Let $Q(n)$ denote “Algorithm A runs in $O(n)$ time”

- $Q(\text{Merge Sort})$: Merge Sort is $O(n \log n)$
FALSE
- $Q(\text{Linear Search})$: Linear Search is $O(n)$
TRUE
- $Q(\text{Binary Search})$: Binary Search is $O(\log n)$
FALSE (it's faster than $O(n)$)

Note: A predicate's truth depends on the specific value assigned!

Compound Predicates - Practice

Let $P(x)$: "x is even", $Q(x)$: " $x > 10$ "

Domain: All integers

Evaluate:

① $P(8) \wedge Q(8) = ?$

② $P(12) \wedge Q(12) = ?$

③ $P(7) \vee Q(7) = ?$

④ $P(15) \rightarrow Q(15) = ?$

Solution: Compound Predicates

$P(x)$: "x is even", $Q(x)$: " $x > 10$ "

① $P(8) \wedge Q(8)$: $\text{TRUE} \wedge \text{FALSE} = \text{FALSE}$

② $P(12) \wedge Q(12)$: $\text{TRUE} \wedge \text{TRUE} = \text{TRUE}$

③ $P(7) \vee Q(7)$: $\text{FALSE} \vee \text{FALSE} = \text{FALSE}$

④ $P(15) \rightarrow Q(15)$: $\text{FALSE} \rightarrow \text{TRUE} = \text{TRUE}$

Remember: Evaluate each predicate first, then apply logical operators!

Universal Quantifier - Practice Set 1

Example 1: Let $P(x)$: " $x^2 \geq x$ "

Domain: All real numbers

Is $\forall x P(x)$ true or false?

Solution: Universal Quantifier Example 1

$P(x)$: " $x^2 \geq x$ ", Domain: All real numbers

Analysis:

We need to check if $x^2 \geq x$ for ALL real numbers.

Rearranging: $x^2 - x \geq 0 \rightarrow x(x - 1) \geq 0$

This is true when:

- $x \leq 0$ or $x \geq 1$

Counterexample: Try $x = 0.5$

$$0.5^2 = 0.25 < 0.5 \quad \text{FALSE}$$

Conclusion: $\forall x P(x)$ is FALSE

Universal Quantifier - Practice Set 2

Example 2: Let $Q(x)$: "If x is prime and $x > 2$, then x is odd"

Domain: All positive integers

Is $\forall x Q(x)$ true or false?

Solution: Universal Quantifier Example 2

$Q(x)$: "If x is prime and $x > 2$, then x is odd"

Analysis:

- The only even prime is 2
- All primes greater than 2 must be odd
- (If they were even and > 2 , they'd be divisible by 2)

Testing:

- $x = 3$: prime, > 2 , odd ✓
- $x = 5$: prime, > 2 , odd ✓
- $x = 7$: prime, > 2 , odd ✓

Conclusion: $\forall x Q(x)$ is **TRUE**

Existential Quantifier - Practice Set 1

Example 1: Let $P(x, y)$: " $x + y = 10$ and $x > y$ "

Domain: All positive integers

Is $\exists x \exists y P(x, y)$ true or false?

Solution: Existential Quantifier Example 1

$P(x, y)$: " $x + y = 10$ and $x > y$ "

We need to find at least one pair (x, y) that satisfies both conditions.

Try: $x = 7, y = 3$

- $7 + 3 = 10$ ✓
- $7 > 3$ ✓

Other valid pairs: $(6, 4), (8, 2), (9, 1)$

Conclusion: $\exists x \exists y P(x, y)$ is **TRUE**

Existential Quantifier - Practice Set 2

Example 2: Let $Q(n)$: “ n is a perfect square and n is prime”

Domain: All positive integers

Is $\exists nQ(n)$ true or false?

Solution: Existential Quantifier Example 2

$Q(n)$: “ n is a perfect square and n is prime”

Analysis:

- Perfect squares: 1, 4, 9, 16, 25, 36, ...
- Primes: 2, 3, 5, 7, 11, 13, ...

Key insight: Any perfect square $n = k^2$ (where $k > 1$) has factors k and k , so it cannot be prime!

What about $n = 1$?

- $1 = 1^2$ is a perfect square
- But 1 is NOT considered prime

Conclusion: $\exists n Q(n)$ is **FALSE**

Translation: English to Logic - Set 1

Example 1: Translate to predicate logic:

“Every programmer knows at least one programming language.”

Step 1: Define predicates and domain

Let:

- $P(x)$: “ x is a programmer”
- $L(x, y)$: “ x knows programming language y ”
- Domain: All people and programming languages

Translation: ?

Solution: Translation Example 1

“Every programmer knows at least one programming language.”

Solution:

$$\forall x(P(x) \rightarrow \exists yL(x, y))$$

Explanation:

- $\forall x$: For every person x
- $P(x) \rightarrow$: If x is a programmer, then...
- $\exists yL(x, y)$: there exists at least one language y that x knows

Note: Use implication ($\rightarrow$) with universal quantifier!

Example 2: Translate to predicate logic:

“Some databases do not support transactions.”

Define predicates:

- $D(x)$: “ x is a database”
- $T(x)$: “ x supports transactions”
- Domain: All software systems

Translation: ?

Solution: Translation Example 2

“Some databases do not support transactions.”

Solution:

$$\exists x(D(x) \wedge \neg T(x))$$

Explanation:

- $\exists x$: There exists at least one system x
- $D(x) \wedge$: such that x is a database AND
- $\neg T(x)$: x does not support transactions

Note: Use conjunction ($\wedge$) with existential quantifier!

Example 3: Translate to predicate logic:
“No student has completed all assignments.”

Define predicates:

- $S(x)$: “ x is a student”
- $A(y)$: “ y is an assignment”
- $C(x, y)$: “ x has completed y ”
- Domain: All people and assignments

Translation: ?

Solution: Translation Example 3

“No student has completed all assignments.”

Solution:

$$\neg \exists x (S(x) \wedge \forall y (A(y) \rightarrow C(x, y)))$$

or equivalently (using De Morgan):

$$\forall x (S(x) \rightarrow \exists y (A(y) \wedge \neg C(x, y)))$$

Second form explanation:

- For every student x
- There exists at least one assignment y
- That x has not completed

Example 1: Translate to English:

$$\forall x(S(x) \rightarrow \exists y(P(y) \wedge E(x, y)))$$

where:

- $S(x)$: “ x is a student”
- $P(y)$: “ y is a project”
- $E(x, y)$: “ x has enrolled in y ”

Translation: ?

Solution: Logic to English Example 1

$$\forall x(S(x) \rightarrow \exists y(P(y) \wedge E(x, y)))$$

Translation:

“Every student has enrolled in at least one project.”

or more verbosely:

“For every person, if that person is a student, then there exists at least one project that the student has enrolled in.”

Example 2: Translate to English:

$$\exists x(C(x) \wedge \forall y(U(y) \rightarrow A(x, y)))$$

where:

- $C(x)$: “ x is a course”
- $U(y)$: “ y is a university”
- $A(x, y)$: “ x is accepted at y ”

Translation: ?

Solution: Logic to English Example 2

$$\exists x(C(x) \wedge \forall y(U(y) \rightarrow A(x, y)))$$

Translation:

“There exists a course that is accepted at every university.”

or:

“Some course is accepted at all universities.”

De Morgan's Laws - Practice Set 1

Example 1: Negate the following statement:
“All students have submitted their homework.”

Original: $\forall x(S(x) \rightarrow H(x))$
where $S(x)$: “ x is a student”, $H(x)$: “ x submitted homework”

Negation: ?

Solution: De Morgan Example 1

Original: $\forall x(S(x) \rightarrow H(x))$

Step-by-step negation:

$$\begin{aligned}\neg \forall x(S(x) \rightarrow H(x)) &\equiv \exists x \neg(S(x) \rightarrow H(x)) \\ &\equiv \exists x \neg(\neg S(x) \vee H(x)) \\ &\equiv \exists x(S(x) \wedge \neg H(x))\end{aligned}$$

English translation:

“There exists at least one student who has not submitted homework.”
or: “Some student has not submitted homework.”

De Morgan's Laws - Practice Set 2

Example 2: Negate the following statement:

“There is a bug in every program.”

Original: $\forall x(P(x) \rightarrow \exists yB(x, y))$

where $P(x)$: “ x is a program”, $B(x, y)$: “ x has bug y ”

Negation: ?

Solution: De Morgan Example 2

Original: $\forall x(P(x) \rightarrow \exists yB(x, y))$

Step-by-step negation:

$$\begin{aligned}\neg \forall x(P(x) \rightarrow \exists yB(x, y)) \\ \equiv \exists x \neg(P(x) \rightarrow \exists yB(x, y)) \\ \equiv \exists x(P(x) \wedge \neg \exists yB(x, y)) \\ \equiv \exists x(P(x) \wedge \forall y \neg B(x, y))\end{aligned}$$

English translation:

“There exists a program that has no bugs.”

or: “Some program is bug-free.”

Nested Quantifiers - Practice Set 1

Example 1: Let $L(x, y)$: “ x loves y ”

Domain: All people

Compare these statements:

① $\forall x \exists y L(x, y)$

② $\exists y \forall x L(x, y)$

What do they mean? Are they equivalent?

Solution: Nested Quantifiers Example 1

$L(x, y)$: “ x loves y ”

Statement 1: $\forall x \exists y L(x, y)$

“Everyone loves someone” (possibly different people)

Statement 2: $\exists y \forall x L(x, y)$

“There is someone whom everyone loves”

Are they equivalent? NO!

- Statement 2 $\rightarrow$ Statement 1 (TRUE)
- Statement 1 $\nrightarrow$ Statement 2 (FALSE)

Order of quantifiers matters!

Nested Quantifiers - Practice Set 2

Example 2: Let $M(x, y)$: “ x has sent a message to y ”

Domain: All users in a system

Translate to predicate logic:

- ① “Every user has sent a message to at least one other user”
- ② “There is a user who has sent messages to all other users”
- ③ “There are two users who have sent messages to each other”

Solution: Nested Quantifiers Example 2

$M(x, y)$: “ x has sent a message to y ”

Solutions:

- ① “Every user has sent a message to at least one other user”

$$\forall x \exists y (x \neq y \wedge M(x, y))$$

- ② “There is a user who has sent messages to all other users”

$$\exists x \forall y (x \neq y \rightarrow M(x, y))$$

- ③ “There are two users who have sent messages to each other”

$$\exists x \exists y (x \neq y \wedge M(x, y) \wedge M(y, x))$$

Domain-Dependent Translations

Example: Translate “Every integer is either even or odd”

Solution 1: Domain = all integers

Let $E(x)$: “ x is even”, $O(x)$: “ x is odd”

$$\forall x(E(x) \vee O(x))$$

Solution 2: Domain = all real numbers

Let $I(x)$: “ x is an integer”

$$\forall x(I(x) \rightarrow (E(x) \vee O(x)))$$

Key point: With universal quantifier, use implication when domain is broader!

Domain-Dependent Translations - Practice

Example: Translate “Some prime number is even”

Solution 1: Domain = all prime numbers

Let $E(x)$: “ x is even”

$$\exists x E(x)$$

Solution 2: Domain = all positive integers

Let $P(x)$: “ x is prime”

$$\exists x (P(x) \wedge E(x))$$

Key point: With existential quantifier, use conjunction when domain is broader!

Example: Express in predicate logic:
“Every user who has admin rights can access all tables”

Define:

- $U(x)$: “ x is a user”
- $A(x)$: “ x has admin rights”
- $T(y)$: “ y is a table”
- $C(x, y)$: “ x can access y ”

Translation: ?

Solution: Database Query Example

“Every user who has admin rights can access all tables”

Solution:

$$\forall x((U(x) \wedge A(x)) \rightarrow \forall y(T(y) \rightarrow C(x, y)))$$

Explanation:

- For every x , if x is a user with admin rights
- Then for every table y
- User x can access table y

Example: Express in predicate logic:

“There exists a path between any two nodes in the graph”

Define:

- $N(x)$: “ x is a node”
- $P(x, y)$: “there is a path from x to y ”

Translation: ?

Solution: Graph Theory Example

“There exists a path between any two nodes in the graph”

Solution:

$$\forall x \forall y ((N(x) \wedge N(y) \wedge x \neq y) \rightarrow (P(x, y) \vee P(y, x)))$$

or if paths are bidirectional:

$$\forall x \forall y ((N(x) \wedge N(y) \wedge x \neq y) \rightarrow P(x, y))$$

Note: This describes a **connected graph**!

Common Mistakes to Avoid

Mistake 1: Using wrong connective with quantifiers

WRONG: "Every student passed" = $\forall x(S(x) \wedge P(x))$

CORRECT: $\forall x(S(x) \rightarrow P(x))$

Mistake 2: Using wrong connective with $\exists$

WRONG: "Some student failed" = $\exists x(S(x) \rightarrow F(x))$

CORRECT: $\exists x(S(x) \wedge F(x))$

Rule of Thumb:

- $\forall$ with $\rightarrow$
- $\exists$ with $\wedge$

Common Mistakes - Negation

Mistake 3: Incorrect negation of quantifiers

Given: $\forall x P(x)$

WRONG: $\neg \forall x P(x) \equiv \forall x \neg P(x)$

CORRECT: $\neg \forall x P(x) \equiv \exists x \neg P(x)$

Mistake 4: Forgetting domain restrictions

“Every positive integer is greater than 0”

If domain = all integers:

WRONG: $\forall x (x > 0)$

CORRECT: $\forall x (x > 0 \rightarrow x > 0)$ or better: specify domain!

Practice Problem Set 1

Translate each statement into predicate logic:

- ① “Not all birds can fly.”
- ② “Every website has at least one administrator.”
- ③ “Some algorithms are both efficient and correct.”
- ④ “No two users have the same email address.”

Try these before moving to solutions!

Solutions: Practice Problem Set 1

Let $B(x)$: “ x is a bird”, $F(x)$: “ x can fly”

$W(x)$: “ x is a website”, $A(x, y)$: “ y is admin of x ”

$G(x)$: “ x is an algorithm”, $E(x)$: “ x is efficient”, $C(x)$: “ x is correct”

$U(x)$: “ x is a user”, $M(x, e)$: “ x has email e ”

Solutions:

① $\neg \forall x (B(x) \rightarrow F(x))$ or $\exists x (B(x) \wedge \neg F(x))$

② $\forall x (W(x) \rightarrow \exists y A(x, y))$

③ $\exists x (G(x) \wedge E(x) \wedge C(x))$

④ $\forall x \forall y ((U(x) \wedge U(y) \wedge x \neq y) \rightarrow \neg \exists e (M(x, e) \wedge M(y, e)))$

Practice Problem Set 2

Translate from predicate logic to English:

① $\forall x(D(x) \rightarrow \exists y(F(y) \wedge S(x, y)))$

where $D(x)$: “ x is a directory”,

$F(y)$: “ y is a file”,

$S(x, y)$: “ x stores y ”

② $\exists x(P(x) \wedge \forall y(P(y) \rightarrow (x = y \vee x < y)))$ where $P(x)$: “ x is prime”

Solutions: Practice Problem Set 2

① $\forall x(D(x) \rightarrow \exists y(F(y) \wedge S(x, y)))$

Translation:

“Every directory stores at least one file.”

② $\exists x(P(x) \wedge \forall y(P(y) \rightarrow (x = y \vee x < y)))$

Translation:

“There exists a smallest prime number.”

or more verbosely:

“There is a prime number that is less than or equal to all other prime numbers.”

(This is true: the answer is 2!)

Advanced Example 1: Multiple Quantifiers

Problem: Express in predicate logic:

“For every student, there is a course that no other student in the class has taken.”

Define:

- $S(x)$: “ x is a student”
- $C(y)$: “ y is a course”
- $T(x, y)$: “student x has taken course y ”
- $CL(x)$: “ x is in this class”

Translation: ?

Solution: Advanced Example 1

“For every student, there is a course that no other student in the class has taken.”

Solution:

$$\forall x((S(x) \wedge CL(x)) \rightarrow \exists y(C(y) \wedge T(x, y) \wedge \\ \forall z((S(z) \wedge CL(z) \wedge z \neq x) \rightarrow \neg T(z, y))))$$

Breaking it down:

- For every student x in this class
- There exists a course y that x has taken
- And for all other students z in the class
- Student z has not taken course y

Advanced Example 2: Symmetry

Problem: Express “The ‘friend’ relationship is symmetric”

Let $F(x, y)$: “ x is a friend of y ”

Domain: All people

Translation: ?

Solution: Advanced Example 2

“The ‘friend’ relationship is symmetric”

Solution:

$$\forall x \forall y (F(x, y) \rightarrow F(y, x))$$

Meaning:

“For any two people, if x is a friend of y , then y is a friend of x .”

Related concepts:

- **Reflexive:** $\forall x F(x, x)$
- **Transitive:** $\forall x \forall y \forall z ((F(x, y) \wedge F(y, z)) \rightarrow F(x, z))$

Real-World Application: Social Networks

Example: Express the following about a social network:

- 1 “Every user has at least one follower”
- 2 “There exists a user who follows everyone”
- 3 “No user follows themselves”

Let $U(x)$: “ x is a user”, $Follows(x, y)$: “ x follows y)’

Solution: Social Network Example

$U(x)$: “ x is a user”, $Follows(x, y)$: “ x follows y ”

Solutions:

- ① “Every user has at least one follower”

$$\forall x(U(x) \rightarrow \exists y(U(y) \wedge Follows(y, x)))$$

- ② “There exists a user who follows everyone”

$$\exists x(U(x) \wedge \forall y(U(y) \wedge x \neq y \rightarrow Follows(x, y)))$$

- ③ “No user follows themselves”

$$\forall x(U(x) \rightarrow \neg Follows(x, x))$$

Real-World Application: E-Commerce

Example: Express the following about an online store:

“Every customer who has placed an order has received at least one confirmation email.”

Define:

- $C(x)$: “ x is a customer”
- $O(x, y)$: “ x has placed order y)’
- $E(x, z)$: “ x has received email z)’
- $Conf(z, y)$: “email z is a confirmation for order y)’

Translation: ?

Solution: E-Commerce Example

“Every customer who has placed an order has received at least one confirmation email.”

Solution:

$$\forall x \forall y ((C(x) \wedge O(x, y)) \rightarrow \exists z (E(x, z) \wedge Conf(z, y)))$$

Explanation:

- For every customer x and order y
- If x has placed order y
- Then there exists an email z
- Such that x received z and z confirms order y

Negation Practice Set

Negate each statement and simplify:

1 $\forall x \exists y (P(x) \rightarrow Q(x, y))$

2 $\exists x \forall y (R(x) \wedge S(y))$

3 $\forall x (P(x) \rightarrow \forall y Q(x, y))$

Apply De Morgan's laws step by step!

Solution: Negation Practice Set

$$① \neg(\forall x \exists y (P(x) \rightarrow Q(x, y)))$$

$$\equiv \exists x \neg \exists y (P(x) \rightarrow Q(x, y))$$

$$\equiv \exists x \forall y \neg (P(x) \rightarrow Q(x, y))$$

$$\equiv \exists x \forall y (P(x) \wedge \neg Q(x, y))$$

$$② \neg(\exists x \forall y (R(x) \wedge S(y)))$$

$$\equiv \forall x \neg \forall y (R(x) \wedge S(y))$$

$$\equiv \forall x \exists y \neg (R(x) \wedge S(y))$$

$$\equiv \forall x \exists y (\neg R(x) \vee \neg S(y))$$

Solution: Negation Practice Set (Continued)

$$3 \quad \neg(\forall x(P(x) \rightarrow \forall yQ(x, y)))$$

$$\equiv \exists x \neg(P(x) \rightarrow \forall yQ(x, y))$$

$$\equiv \exists x(P(x) \wedge \neg \forall yQ(x, y))$$

$$\equiv \exists x(P(x) \wedge \exists y \neg Q(x, y))$$

Key steps:

- $\neg \forall$ becomes $\exists \neg$
- $\neg \exists$ becomes $\forall \neg$
- $\neg(p \rightarrow q) \equiv p \wedge \neg q$

Challenge Problem 1

Express in predicate logic:

“Every non-empty set has a smallest element” (Well-ordering principle for finite sets)

Define:

- $Set(S)$: “ S is a set”
- $Empty(S)$: “ S is empty”
- $In(x, S)$: “ x is in S ”
- $x \leq y$: “ x is less than or equal to y ”

Translation: ?

Solution: Challenge Problem 1

“Every non-empty set has a smallest element”

Solution:

$$\forall S((Set(S) \wedge \neg Empty(S)) \rightarrow \\ \exists x(In(x, S) \wedge \forall y(In(y, S) \rightarrow x \leq y)))$$

Explanation:

- For every set S that is non-empty
- There exists an element x in S
- Such that for all elements y in S
- x is less than or equal to y

Challenge Problem 2

Express in predicate logic:

"If a function is continuous at every point, then it is continuous."

Define:

- $F(f)$: " f is a function"
- $P(x)$: " x is a point in the domain"
- $ContAt(f, x)$: " f is continuous at point x)"
- $Cont(f)$: " f is continuous"

Translation: ?

Solution: Challenge Problem 2

“If a function is continuous at every point, then it is continuous.”

Solution:

$$\forall f (F(f) \wedge \forall x (P(x) \rightarrow \text{ContAt}(f, x)) \rightarrow \text{Cont}(f))$$

Explanation:

- For every function f
- If f is continuous at every point x in its domain
- Then f is continuous

Note: This is actually the definition of continuity!

Summary: Key Takeaways

Remember these key points:

① Quantifier + Connective Rules:

- $\forall$ with $\rightarrow$: $\forall x(P(x) \rightarrow Q(x))$
- $\exists$ with $\wedge$: $\exists x(P(x) \wedge Q(x))$

② De Morgan's Laws:

- $\neg\forall xP(x) \equiv \exists x\neg P(x)$
- $\neg\exists xP(x) \equiv \forall x\neg P(x)$

③ Order matters in nested quantifiers!

④ Always specify domain clearly

⑤ Practice translating both ways: English Logic

Practice Makes Perfect!

Tips for mastering predicate logic:

- Practice translating everyday statements
- Draw truth tables for small domains
- Check your translations by converting back to English
- Learn to recognize common patterns
- Always verify with concrete examples

Keep practicing and you'll master it!